

# William Zhu

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## Professional Experience

### Northwestern Mutual

#### *Software Engineer | June 2025 - Present*

- Develop full-stack software for enterprise insurance systems across Java backend services, React interfaces, APIs, databases, cloud infrastructure, testing, and production support.
- Contributed to backend and frontend components for an internal policy reporting platform, supporting data retrieval, display logic, testing, and production debugging.
- Owned early backend development for a customer-facing payment history feature, setting up the service repository and building API endpoints that structured policy and payment data for frontend display.
- Adapted the feature through evolving architecture changes, including REST APIs, GraphQL workflows, and a serverless AWS Lambda design with Amazon S3 and HashiCorp Vault integration.
- Identified requirement and design gaps, proposed clearer API response patterns, and supported releases through CI/CD, regression testing, Grafana-based debugging, and incident resolution.

### Nuu Mobile

#### *Machine Learning Researcher | May 2025 - Present*

- Conduct applied machine learning research on churn prediction using sparse, imbalanced, real-world telecom and device data; first author of an academic manuscript currently under review.
- Implemented and benchmarked 11 machine learning, deep learning, and ensemble models across stratified and time-aware validation settings.
- Evaluated model performance using F2-score, F3-score, ROC-AUC, threshold tuning, SMOTE, and SMOTETomek, analyzing how temporal drift and validation strategy affect model selection.

#### *Full Stack Software Engineer | UW-Madison x Nuu Mobile Client Project | January 2025 - May 2025*

- Led a team of 4 in building a full-stack churn prediction platform with Flask REST APIs, PostgreSQL, data preprocessing, prediction workflows, and customer analytics.
- Containerized the application with Docker, deployed it on AWS EC2 for stakeholder access, and worked directly with company stakeholders through demos and design iterations.

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## Projects

### ContextGuard

#### *Creator / Software Engineer | June 2026*

- Built a VS Code extension that helps developers inspect AI coding context, estimate token footprint, and understand what files and configuration may be loaded into assistant workflows.
- Implemented workspace scanning, instruction-file detection, context preview/bundling, budget indicators, and sensitive-file detection for AI-assisted development.

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## Education

**University of Wisconsin-Madison:** Honors College of Letters and Science: September 2021 - May 2025

3.87/4.0 GPA - 6x Dean's List Award Winner & Dean's Prize Finalist

Bachelor of Science in Computer Science with a Minor in Business

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## Skills

**Languages:** Java, Python, TypeScript/JavaScript, SQL, C, C++

**Backend & Cloud:** Flask, REST APIs, GraphQL, AWS, Docker, Kubernetes, GitLab CI/CD, HashiCorp Vault

**Frontend & Tools:** React, React Native, Postman/Bruno, Grafana, Git, Agile Development, Code Reviews

**ML & AI:** scikit-learn, PyTorch, Optuna, Deep Learning, LLMs, F2/F3-score, ROC-AUC, Time-Aware CV, SMOTE

**Quality & Ops:** Automated and Regression Testing, Release Validation, Production Debugging, Technical Documentation

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Passionate about clean code, elegant designs, and occasionally lets the cat commit a few lines